# Grid readiness to 2040

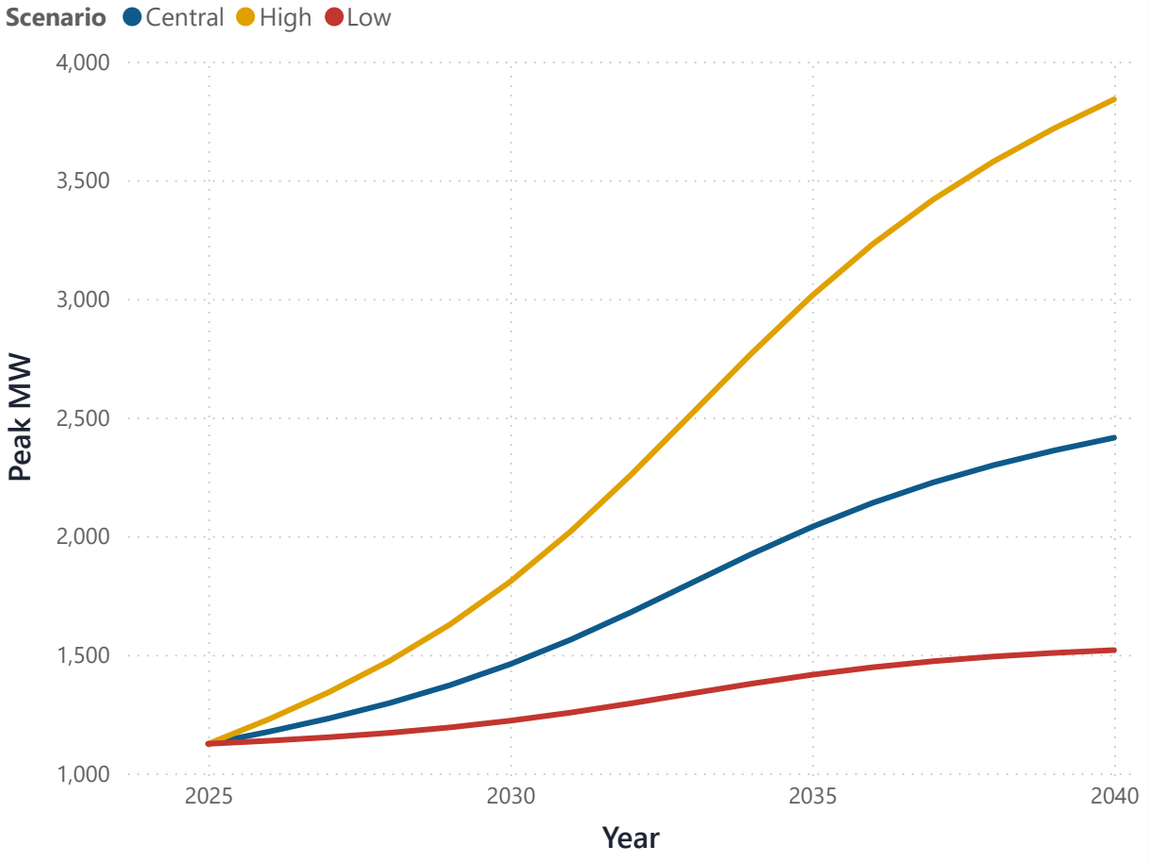
Cork and Dublin demand substations · winter peak · public CSO, Census and ESB Networks data

County: Cork | Dublin

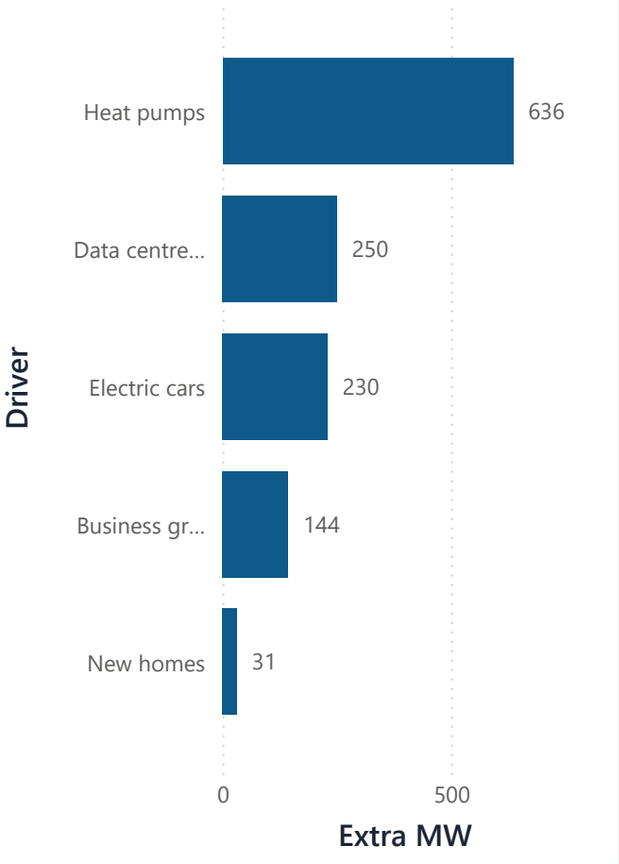
Model scenario: Central | High | Low

| 1,124 | 2,414 | 114.7% | 77 | 21 | €379m – €817m |
|---|---|---|---|---|---|
| Winter peak 2025 (MW) | Winter peak 2040 (MW) | Growth to 2040 | Stations short by 2040 | Need a new station | Station investment (P10–P90) |

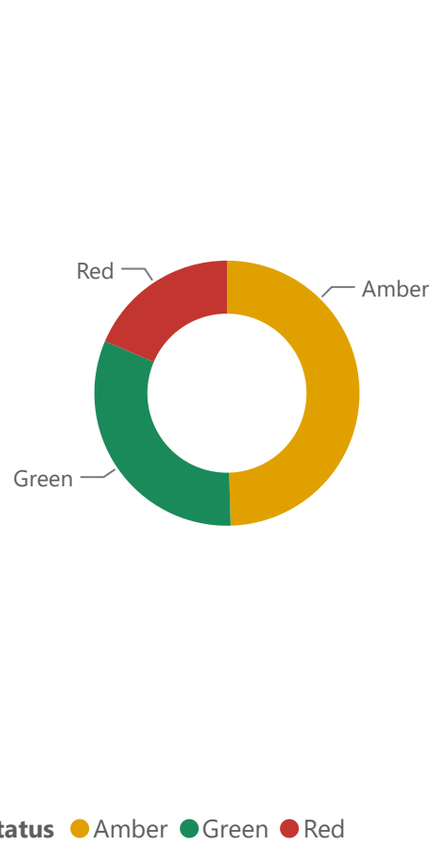
## Winter peak demand by model scenario (MW)




## What adds demand by 2040 (MW)




## Station status in 2040 (model)



# Readiness: ESB Networks' own forecast

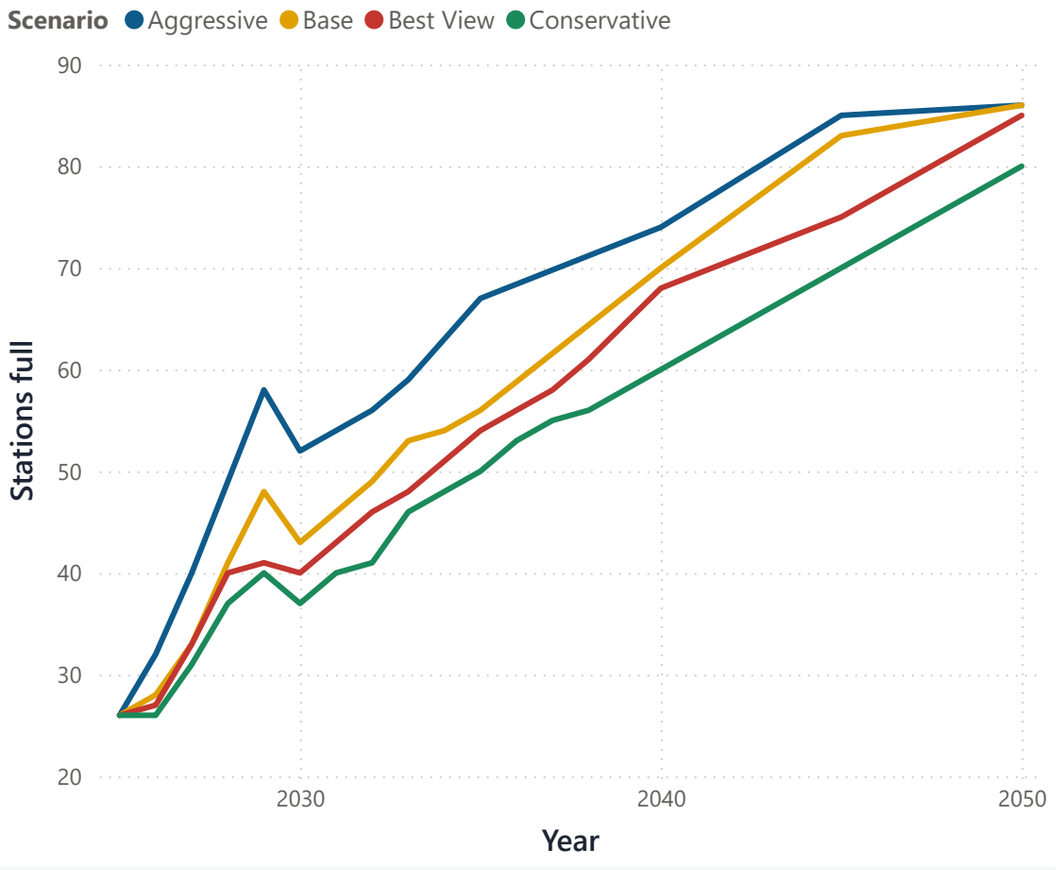
Spare capacity at each station under ESB Networks' four scenarios, against my model's 2040 result

County: Cork | Dublin

ESB Networks scenario: Base

| 77 | 70 | 113 | 2025 |
|---|---|---|---|
| My model: stations short by 2040 | ESB Networks: stations full by 2040 | Demand stations | First year a station is full |

## Stations with no spare capacity, all four ESB Networks scenarios




## Spare capacity by station (MW): red = full, amber = under 5 MW

| Station | 2025 | 2030 | 2035 | 2040 | 2045 | 2050 |
|---|---|---|---|---|---|---|
| Adamstown | 18.1 | 15.0 | 25.6 | 10.5 | 8.1 | 5.8 |
| Artane | 13.4 | 9.0 | 11.5 | 2.5 | 0.0 | 0.0 |
| Balbriggan | 11.9 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Balgaddy | 3.1 | 2.5 | 0.5 | 0.0 | 0.0 | 0.0 |
| Ballinclea | 4.0 | 3.0 | 3.9 | 0.9 | 0.0 | 0.0 |
| Ballyboden | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Ballycoolin | 5.4 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Ballymount | 0.2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Ballymun | 6.4 | 4.2 | 5.1 | 0.9 | 0.0 | 0.0 |
| Batter Lane | 0.0 | 41.3 | 82.6 | 41.3 | 41.3 | 41.3 |
| Bedford Row | 14.8 | 13.0 | 22.7 | 9.6 | 7.8 | 6.0 |
| Belfield | 3.6 | 1.1 | 2.5 | 0.0 | 0.0 | 0.0 |
| Belgard | 2.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Belmayne | 0.0 | 17.2 | 34.4 | 17.2 | 17.2 | 17.2 |
| Blackrock | 36.4 | 24.1 | 30.7 | 3.3 | 0.0 | 0.0 |
| Blundelstown | 0.0 | 86.1 | 159.5 | 79.7 | 79.7 | 79.7 |
| Brewery Road | 2.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Cabra | 6.4 | 0.3 | 0.0 | 0.0 | 0.0 | 0.0 |
| Camden Row | 1.7 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Carrickmines | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| Castleknock | 3.3 | 2.3 | 3.0 | 0.7 | 0.0 | 0.0 |
| Central Park | 17.9 | 15.3 | 25.6 | 10.2 | 7.7 | 5.1 |

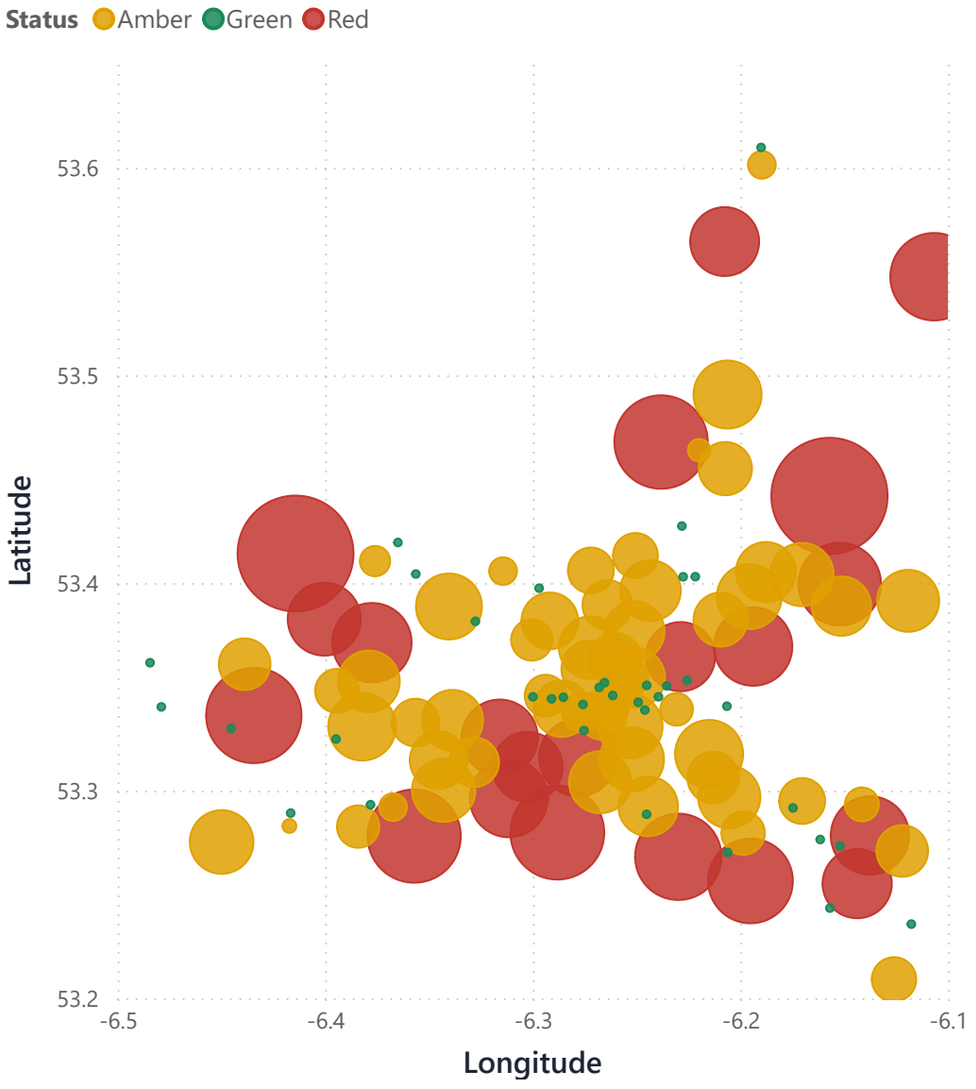
# Site scorecard

Which stations run short, by how much, from which year, and what they need

County: Cork | Dublin

Action needed: New station | No action | Upgrade

## Where the short stations are (bubble = 2040 shortfall)




## Station scorecard (sorted by shortfall)

| Station | kV | Spare now (MW) | Extra by 2040 (MW) | Shortfall (MW) | Action | Full from |
|---|---|---|---|---|---|---|
| Malahide | 38 | 0.0 | 32.4 | 32.4 | New station | 2026 |
| Macetown | 110 | 0.0 | 32.2 | 32.2 | New station | 2026 |
| Balgaddy | 38 | 3.0 | 24.1 | 21.1 | New station | 2028 |
| Ballyboden | 38 | 0.0 | 20.4 | 20.4 | New station | 2026 |
| Glasmore | 38 | 0.0 | 20.2 | 20.2 | New station | 2026 |
| Oldbawn | 38 | 0.0 | 20.1 | 20.1 | New station | 2026 |
| Loughshinny | 38 | 0.8 | 18.2 | 17.5 | New station | 2027 |
| Sandyford | 38 | 0.0 | 17.0 | 17.0 | New station | 2026 |
| Carrickmines | 38 | 0.0 | 16.3 | 16.3 | New station | 2026 |
| Grange (Dr) | 38 | 0.0 | 15.4 | 15.4 | New station | 2026 |
| Castleknock | 38 | 3.2 | 17.3 | 14.1 | New station | 2029 |
| Clontarf | 38 | 0.4 | 14.2 | 13.7 | New station | 2026 |
| Sallynoggin Road | 38 | 0.8 | 14.5 | 13.7 | New station | 2027 |
| Templeogue | 38 | 4.0 | 17.6 | 13.6 | New station | 2030 |
| Garville Avenue | 38 | 0.6 | 13.9 | 13.4 | New station | 2027 |
| Crumlin | 38 | 5.2 | 18.2 | 12.9 | New station | 2031 |
| Coolmine | 38 | 0.0 | 11.7 | 11.7 | New station | 2026 |
| Kimmage | 38 | 0.0 | 10.7 | 10.7 | New station | 2026 |
| Fairview | 38 | 1.4 | 11.8 | 10.4 | New station | 2028 |
| Loughlinstown | 38 | 0.4 | 10.8 | 10.4 | New station | 2026 |
| Courtlough | 38 | 0.0 | 10.3 | 10.3 | New station | 2026 |
| Batter Lane | 110 | 0.0 | 10.0 | 10.0 | Upgrade | 2026 |
| Nutley Lane | 110 | 0.0 | 10.0 | 10.0 | Upgrade | 2026 |
| Yellowmeadows | 110 | 0.0 | 9.9 | 9.9 | Upgrade | 2026 |
| Dunsink | 110 | 0.0 | 9.3 | 9.3 | Upgrade | 2026 |
| Coolock | 38 | 5.6 | 14.6 | 9.0 | Upgrade | 2032 |